

Data Flow Diagram (DFD) Components

Project Name: CaptionAI: AI-powered social media caption generator

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Maximum Marks: 2 Marks

1. DFD Component Mapping

Component Type	Name	Description / Role in CaptionAI
External Entity	User (Entrepreneur / Content Creator)	Sends text inputs, product details, and platform preferences; receives generated captions and hashtags.
External Entity	LLM API Provider (e.g., OpenAI / Anthropic)	External artificial intelligence system that processes the prompt engineered data and returns generated text variants.
Process 1	1. Request Handling	Validates user input text, formats payload parameters, and handles user authentication.
Process 2	2. Content Generation & Processing	Construct prompts with platform constraints (e.g., Instagram/LinkedIn rules), pushes them to the external LLM API, and parses responses.
Process 3	3. Export & Analytics Management	Tracks click/copy actions, manages user history logs, and triggers notification alerts.
Data Store	User & Request Datastore	Retains user profile metadata, history records, and generated caption logs for later dashboard reference.

2. System Data Flow Matrix

Source	Target	Data Flow Label (What data is being passed)
User	1. Request Handling	Product text input, platform preference, and account tokens.
1. Request Handling	User	Input validation acknowledgment or login validation status.
1. Request Handling	User & Request Datastore	Validated request parameters and temporary profile session data.
User & Request Datastore	2. Content Generation & Processing	Saved brand voice preferences and target platform formatting criteria.
2. Content Generation & Processing	LLM API Provider	Structured prompt package (payload data).
LLM API Provider	2. Content Generation & Processing	Raw generated response strings containing text variations.
2. Content Generation & Processing	3. Export & Analytics Management	Clean text output variants, single-symbol hashtags, and CTAs.
3. Export & Analytics Management	User & Request Datastore	Log records of successfully generated text structures for user history.
3. Export & Analytics Management	User	Final platform-optimized text package (ready to copy).

